Physics A (H156, H556)

Unified - Electromagnetism ppq
KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:
Additional resources may be supplied with this paper.

Other materials required:
- Pencil
- Ruler (cm/mm)

Duration: Not set

Candidate forename		Candidate surname	

Centre number						Candidate number				

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets **[ ]** at the end of each question or part question.
- The total number of marks for this paper is **101**.
- The total number of marks may take into account some 'either/or' question choices.

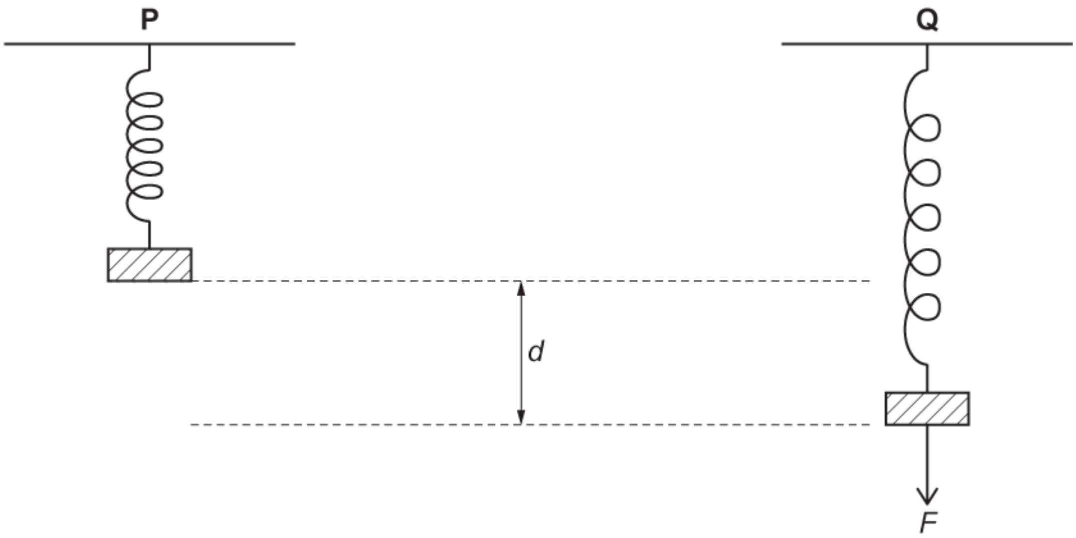
1(a) The length of an unloaded spring is approximately 4 cm.

The force constant k of the spring is 0.62 N cm^{-1}.

The figure below shows a block of mass 0.20 kg attached to one end of the spring. The other end of the spring is attached to a fixed support vertically above the block.

In position **P** the block rests in equilibrium. The extension of the spring is 3.2 cm.

In position **Q** a downwards force F has been applied to the block, so that it now rests a distance d below its position at **P**. The extension of the spring is now 8.5 cm.



The force F is removed.

i. Calculate the magnitude of the block's initial acceleration at the instant that the force F is removed.

Assume that the spring is not extended beyond its limit of proportionality.

acceleration = ... m s^{-2} **[3]**

ii. The block now moves with simple harmonic motion.

Calculate the frequency of this motion.

frequency = ... Hz **[3]**

(b) The block is replaced by a strong magnet **L** of slightly greater mass.

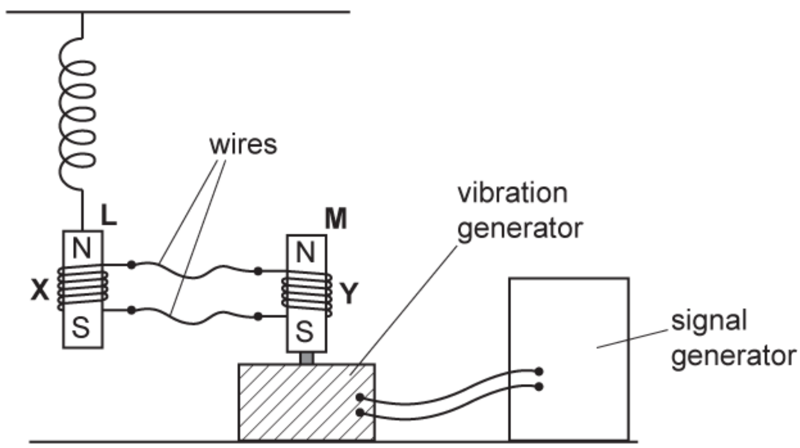
The oscillation frequency of this new arrangement is 2.5 Hz.

The magnet **L** is placed inside a coil **X** of insulated copper wire.

The coil **X** is connected with long wires to a second, identical coil **Y**.

A second strong magnet **M** is placed inside **Y** and attached to a vibration generator.

The vibration generator is then forced to oscillate with a frequency of approximately 2.5 Hz by adjusting the signal generator.



i. As magnet **M** oscillates, it moves in and out of coil **Y**.

The magnet **L** also begins to oscillate.

Explain why **L** oscillates.

---

[3]

ii. The frequency of the vibration generator is now varied between 0.5 Hz and 5.0 Hz.

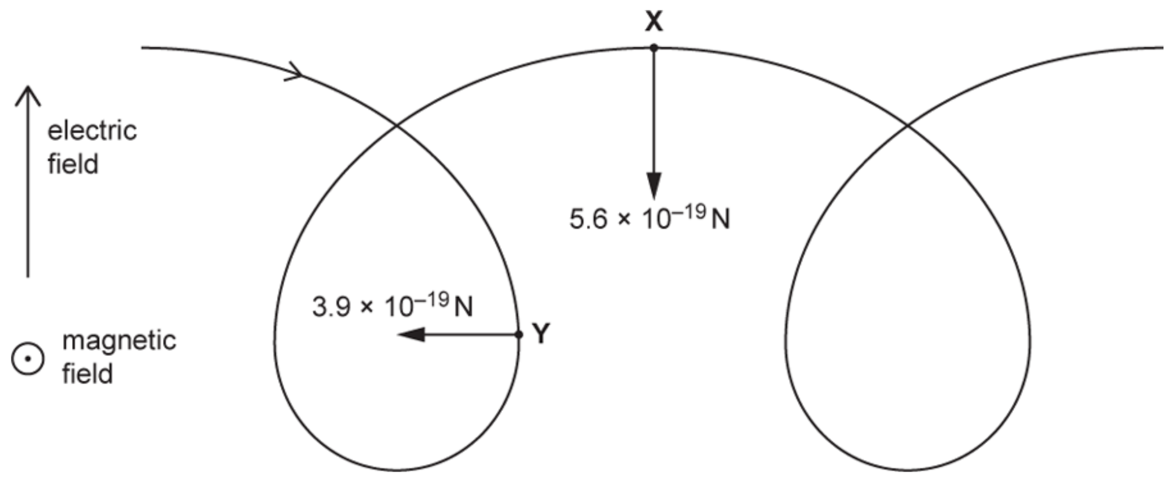
Suggest how the amplitude and frequency of the oscillations of **L** will change as the frequency of the generator is varied.

You may draw a diagram to support your answer.

[3]

2(a) The figure below shows the path of a proton moving in a region occupied by both an electric field and a magnetic field.

The direction of the electric field lines is perpendicular to the direction of the magnetic field lines.



The uniform electric field is directed upwards, with electric field strength $E = 0.90\ \text{N C}^{-1}$.

The uniform magnetic field is directed out of the plane of the paper, with magnetic flux density $B = 5.0 \times 10^{-5}$ T.

At point **X** the proton is moving horizontally to the right. The magnitude of the **magnetic** force at **X** is 5.6×10^{-19} N.

At point **Y** the proton is moving vertically downwards. The magnitude of the **magnetic** force at **Y** is 3.9×10^{-19} N.

The **electric** forces acting on the proton at **X** and **Y** are **not** shown in the figure.

Show that the magnitude of the constant **electric** force acting on the proton is about 10^{-19} N.

[1]

(b)

i. Suggest why the **magnetic** force acting on the proton has a different magnitude at **X** than at **Y**.

[1]

ii. At **X**, the motion of the proton is instantaneously equivalent to motion in a circle at a constant speed.

Calculate the radius of this circular motion.

radius = ... m **[4]**

iii. **1** Calculate the magnitude of the resultant force on the proton at **Y**.

resultant force = ... N **[2]**

2 Explain why the motion of the proton at **Y** is **not** instantaneously equivalent to motion in a circle at a constant speed.

[2]

3

*A student, supervised by their teacher, carries out an experiment with three unlabelled radioactive sources.

The student is told that each source emits only one type of radiation. One emits gamma rays, one emits beta-minus particles and one emits beta-plus particles.

The student has the following equipment:
- a selection of materials with different thicknesses
- a strong magnet
- a radiation counter (GM tube and counter).

Explain how the student can use this equipment to determine safely which radiation each source emits.

You may use the space below to draw a diagram.

[6]

4 This question is about an electric cooker, which consists of an oven and an electromagnetic induction hob.

The electromagnetic induction hob is shown in **Fig. 4.1.**

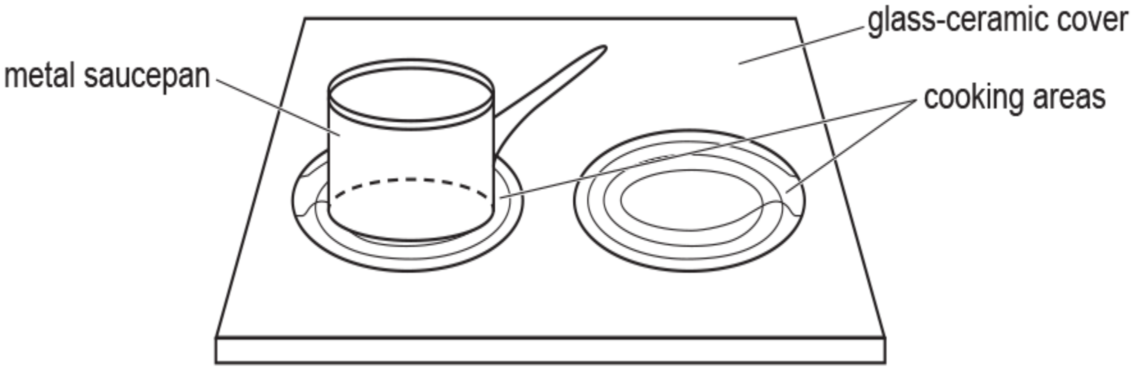


Fig. 4.1

Each cooking area has a coil below the glass-ceramic cover. When switched on, the coils carry a high-frequency **alternating** current.

A metal saucepan is placed above one of the coils. A large alternating current is induced in the saucepan base, and this causes the saucepan to heat up.

i. **Fig. 4.2**shows one of the coils at a time when the current is in the direction indicated by the arrows.

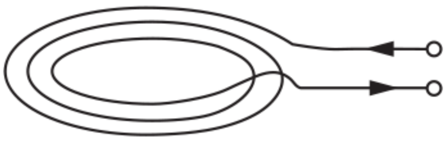


Fig. 4.2

On **Fig. 4.2**, sketch the magnetic field pattern for the current-carrying coil. **[2]**

ii. Fig. 4.3 shows the path of the large alternating current induced in the metal base of the saucepan

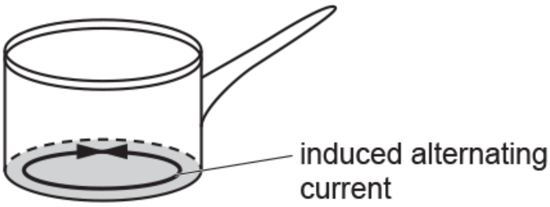


Fig. 4.3

Explain the origin of this large current.

--- [2]

iii. Explain why it would be safe for a person to place a hand on the cooking area before the saucepan is put onto it.

--- [2]

5 Fig. 3.1 shows the design of a 'mechanical' torch.

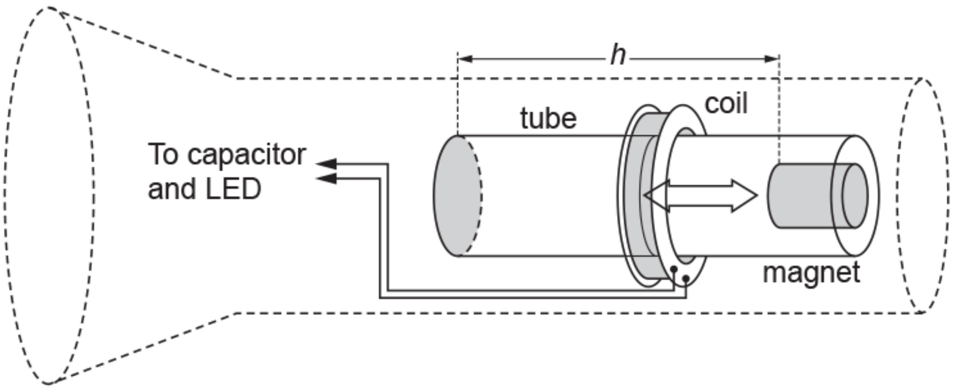


Fig. 3.1

There is no battery in the torch. Instead, when the torch is inverted, the magnet falls a short vertical distance *h* through the coil of wire, as shown in Fig. 3.2. This induces an electromotive force (e.m.f.) across the ends of the coil. The e.m.f. is used to store charge in a capacitor, which lights a light-emitting diode (LED) when it discharges.

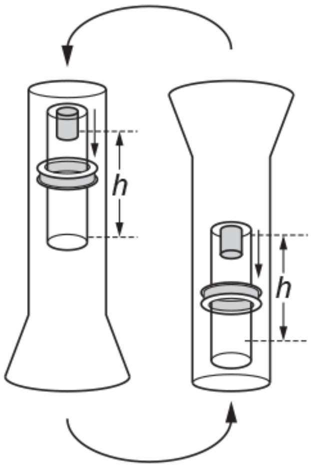


Fig. 3.2

Fig. 3.3 shows the variation with time of the e.m.f. generated as the magnet falls the distance *h.*

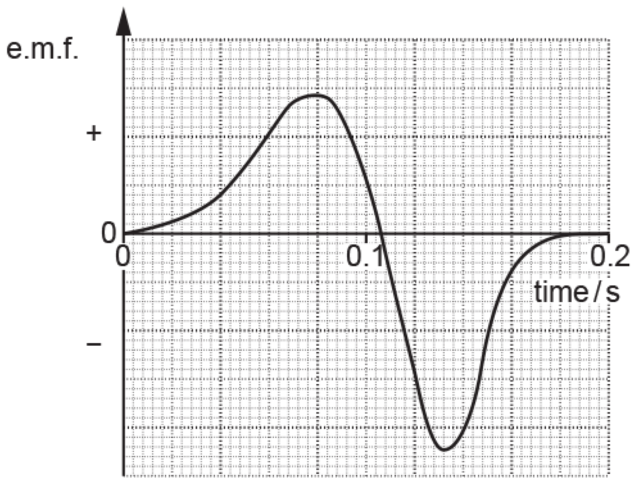


Fig. 3.3

Explain the shape of the curve in Fig. 3.3.

[3]

6(a)

A magnet rotates inside a shaped soft iron core. A coil is wrapped around the iron core as shown in Fig. 5.1. The coil is connected to an oscilloscope.

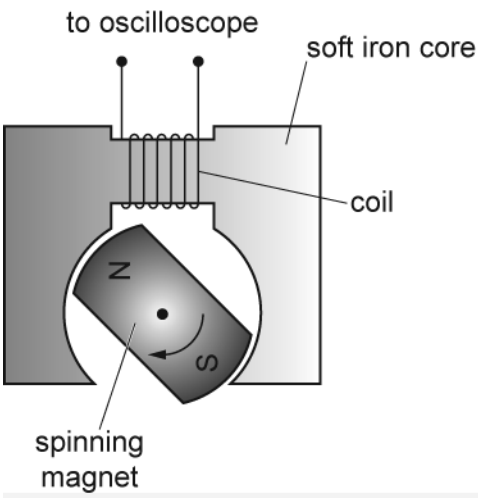


Fig. 5.1

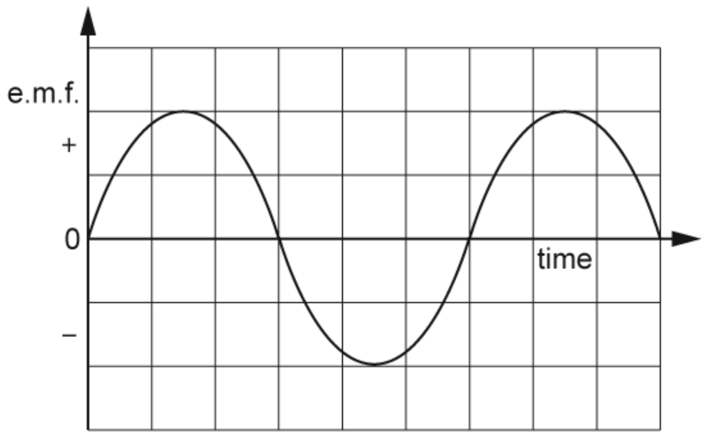


Fig. 5.2

The spinning magnet induces an e.m.f. in the coil. A graph of the e.m.f. displayed on the oscilloscope screen is shown in Fig. 5.2.

i. Explain the shape of the graph in terms of the magnetic flux linking the coil.

-- [2]

ii. On Fig. 5.3 sketch a graph of the magnetic flux linkage of the coil against time. The variation of the induced e.m.f. across the coil is shown as a dotted line.

[1]

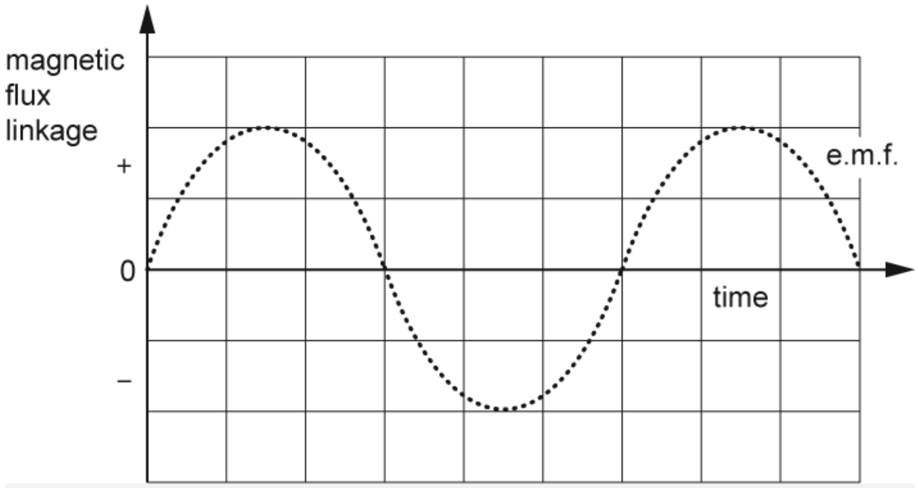


Fig. 5.3

iii. The coil shown in Fig. 5.1 has 150 turns. The maximum induced e.m.f. V_0 across the coil is 1.2 V when the magnet is rotating at 24 revolutions per second.

Calculate the maximum **magnetic flux** through the coil using the equation

$$V_0 = 2\pi \times (\text{frequency}) \times (\text{maximum magnetic flux linkage})$$

Give a unit with your answer.

maximum flux = -------------------- unit ------------ **[2]**

(b)

A student is given a transformer with coils **X** and **Y**, as shown in Fig. 5.4.

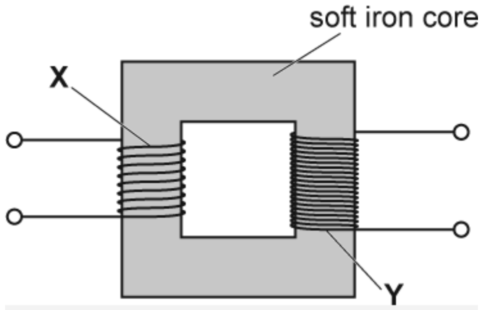


Fig. 5.4

The student is intending to investigate how the maximum induced e.m.f. V_0 in coil **Y** depends on the frequency f of the alternating current in coil **X**.

The changing magnetic flux density in coil **X** induces an e.m.f. in coil **Y**. Faraday's law indicates that the maximum induced e.m.f. V_0 should be directly proportional to f.

Describe how you would investigate the suggested relationship between V_0 and f in the laboratory using these coils. In your description include all of the equipment used and how you would analyse the data collected.

Use the space below to draw a suitable diagram.

[6]

7(a) A small thin rectangular slice of semiconducting material has width a and thickness b and carries a current I. The current is due to the movement of electrons. Each electron has charge $-e$ and mean drift velocity v.
A uniform magnetic field of flux density B is perpendicular to the direction of the current and the top face of the slice as shown in Fig. 2.1.

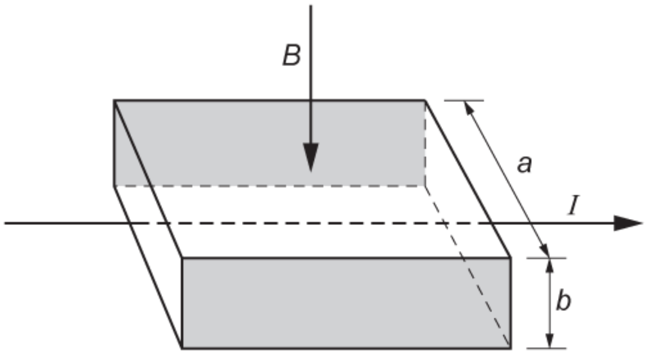


Fig. 2.1

As soon as the current is switched on, the moving electrons in the current are forced towards the shaded rear face of the slice where they are stored. This causes the shaded faces to act like charged parallel plates. Each electron in the current now experiences both electric and magnetic forces. The resultant force on each electron is now zero.

Write the expressions for the electric and magnetic forces acting on each electron and use these to show that the magnitude of the potential difference V between the shaded faces is given by

$$V = Bva.$$

[3]

(b) Here are some data for the slice in a particular experiment.

number of conducting electrons per cubic metre, $n = 1.2 \times 10^{23}\ m^{-3}$

$a = 5.0$ mm

$b = 0.20$ mm

$I = 60$ mA

$B = 0.080$ T

Use this data to calculate

i. the mean drift velocity v of electrons within the semiconductor

v = ____________________ $m\ s^{-1}$ **[3]**

ii. the potential difference V between the shaded faces of the slice.

V = ____________________ V **[1]**

(c) The slice is mounted and used as a measuring instrument called a Hall probe.
A cell is connected to provide the current in the slice. The potential difference across the slice is measured by a separate voltmeter.

A student wants to measure the magnetic flux density between the poles of two magnets mounted on a steel yoke as shown in Fig. 2.2. The magnitude of the flux density is between 0.02 T and 0.04 T.

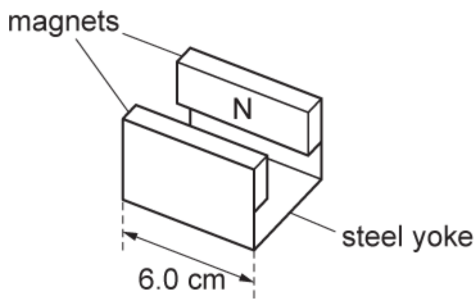


Fig. 2.2

i. Suggest **one** reason why this Hall probe is **not** a suitable instrument to measure the magnetic flux density for the arrangement shown in Fig. 2.2.

--- **[1]**

ii. Another method of measuring the magnetic flux density for the arrangement shown in Fig. 2.2 is to insert a current-carrying wire between the poles of the magnet.
Explain how the magnetic flux density can be determined using this method and discuss which measurement in the experiment leads to the greatest uncertainty in the value for the magnetic flux density.

---

[4]

8(a) * A student is to investigate the magnetic field inside and around a solenoid.

It is suggested that the magnetic field strength *B* inside a long solenoid is determined by various quantities,

namely $B \propto \frac{NI}{L}$

where *N* is the number of turns, *L* is the length of the solenoid and *I* is the current in the wire.

Apparatus is set up for an experiment as shown in Figure 6.1.

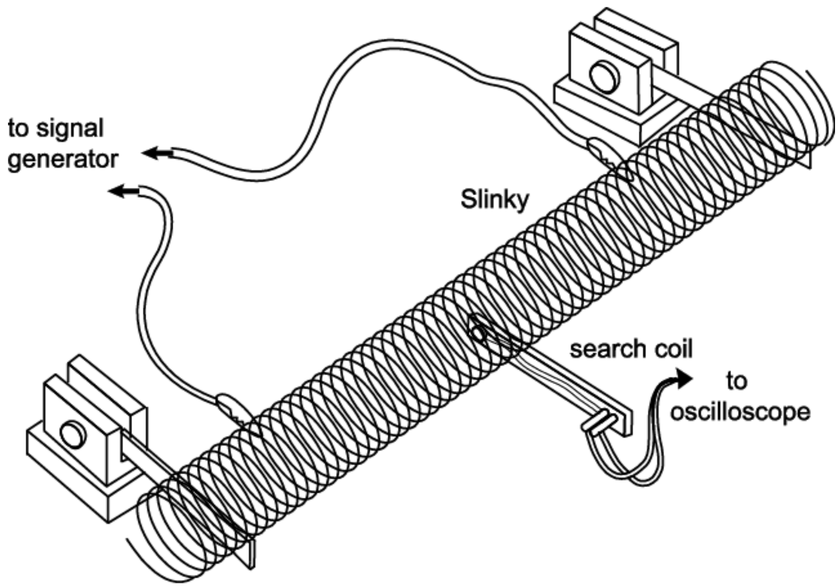


Fig. 6.1

A Slinky is a long spring about 70 mm in diameter which can be stretched easily and uniformly. The search coil has 5000 turns and the signal generator can produce a constant alternating current at a frequency between 0 and 1 kHz.

Plan an experiment using this equipment to investigate the validity of the relationship between *B,* at the centre of the solenoid, and **one** of the variables *N* or *L*. Explain how you will make your measurements, how sensitive they will be and the steps that you will take to make this a valid test.

[6]

(b) Fig. 6.2 shows a soft iron ring of variable circular cross-section. It has four coils containing 2, 3, 4 and 5 turns wound around it. The cross-sectional area of the ring is different for each coil.

A cell is connected across the coil with three turns.

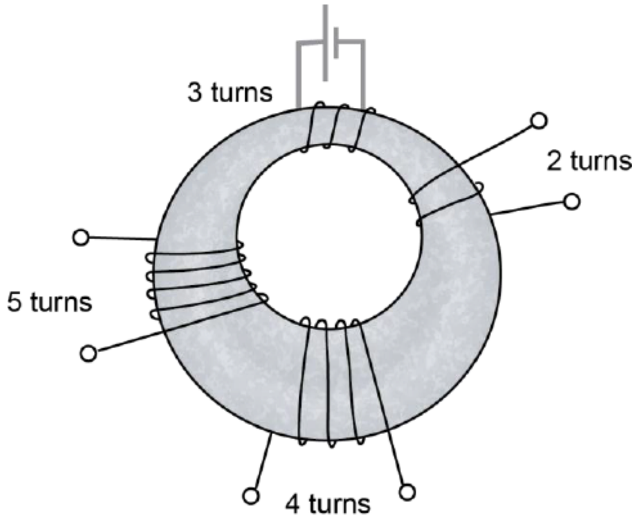


Fig. 6.2

i. Draw on Fig. 6.2 the complete paths of **two** lines of magnetic flux produced by the three-turn coil when there is a current in it.

[1]

ii. State which **one** of the following three quantities,

magnetic flux *magnetic flux density* *magnetic flux linkage*

is most nearly the same for all four coils in Fig. 6.2. Give a reason for your answer.

--- **[1]**

iii. Write down **one** of the **other** two quantities in **(ii)** above. State in which coil this quantity has the largest value. Give a reason for your answer.

--- **[2]**

9(a) Fig. 5.1 shows a horizontal copper wire placed between the opposite poles of a permanent magnet. The wire is held in tension *T* by the clamps at each end. The length of the wire in the magnetic field of flux density 0.032 tesla is 6.0 cm.

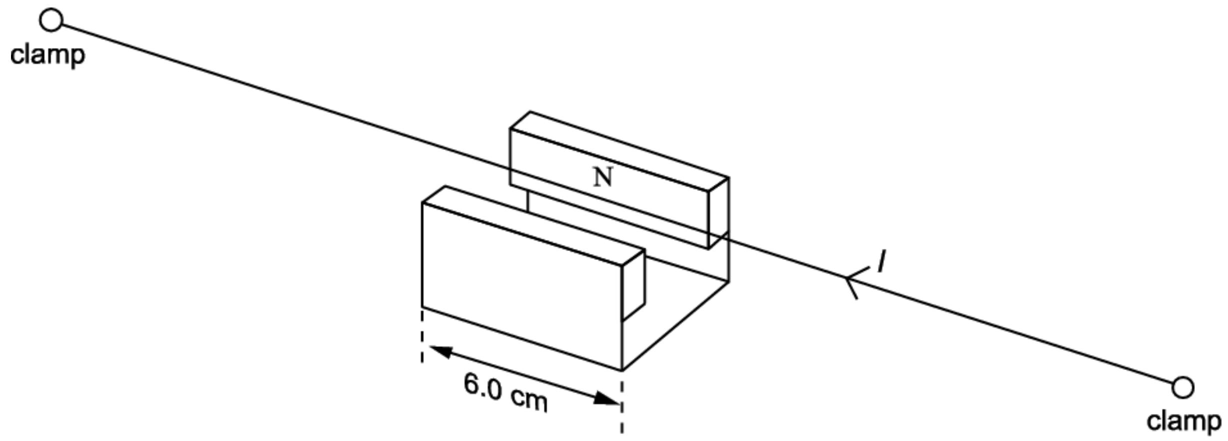


Fig. 5.1

A direct current *I* of 2.5 A is passed through the wire as shown.

i. On Fig. 5.1 draw an arrow to indicate the direction of the force *F* on the wire.

[1]

ii. Calculate the magnitude of *F*.

F = ____________________ N **[1]**

(b) The direct current is changed to an alternating current of constant amplitude and variable frequency, causing the wire to oscillate. The frequency of the current is increased until the fundamental natural frequency of the wire is found as shown in Fig. 5.2. This is 70 Hz.

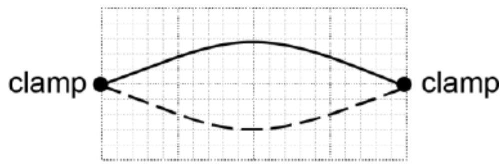


Fig. 5.2

i. In the situation shown in Fig. 5.2 the amplitude of the oscillation of the centre point of the wire is 4.0 mm. Calculate the maximum acceleration of the wire at this point.

maximum acceleration = ____________________ m s^{-2} **[2]**

ii. The frequency is increased until another stationary wave pattern occurs. The amplitude of this stationary wave is much smaller.

1. Sketch this pattern on Fig. 5.3 and state the frequency

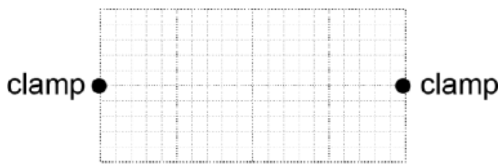


Fig. 5.3

frequency = ____________________ Hz **[1]**

2. Explain why the amplitude is so small. Suggest how the experiment can be modified to increase the amplitude.

____

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____

____

____ **[3]**

(c)

The speed v of a transverse wave along the wire is given by $v = \sqrt{\frac{T}{\mu}}$ where T is the tension and μ is the mass per unit length of the wire.

i. Assume that both the length and mass per unit length remain constant when the tension in the wire is halved. Calculate the frequency of the new fundamental mode of vibration of the wire.

frequency = ____________________ Hz **[1]**

ii. In practice the mass per unit length changes because the wire contracts when the tension is reduced. For the situation in which the tension is halved the strain reduction is found to be 0.4%.

1. Calculate the percentage change in μ. State both the size and sign of the change.

percentage change in μ = ____________________ % **[1]**

2. Write down the percentage error this causes in your answer to **(i)**. State, giving your reasoning, whether the actual frequency would be higher or lower than your value.

____

____

____ [2]

10(a) * **Fig. 5.1** shows a simple a.c. generator being tested by electrical engineers.

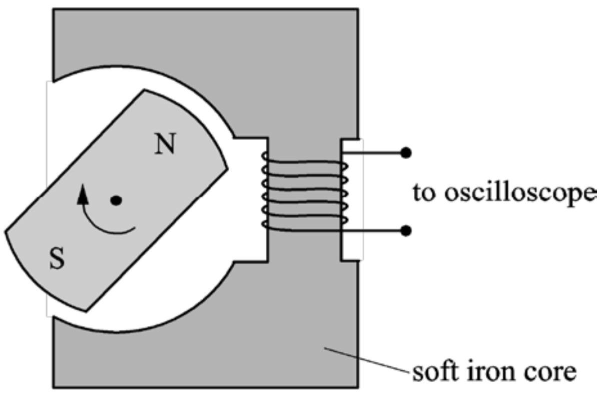

Fig. 5.1

It consists of a magnet, on the shaft of a **variable speed** motor, being rotated inside a cavity in a soft iron core. The output from the coil, wound on the iron core, is connected to an oscilloscope. The grid of **Fig. 5.2** shows a typical output voltage which would be displayed on the oscilloscope screen.

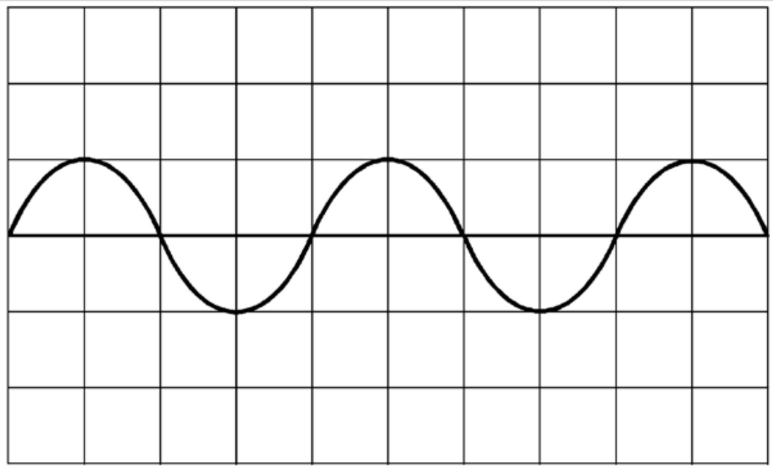

Fig. 5.2

According to Faraday's law the e.m.f. induced is directly proportional to the rate of change of flux linkage. In the context of this experiment, the maximum e.m.f. induced is directly proportional to the frequency of rotation of the magnet.

Use the apparatus above to plan an experiment to validate Faraday's law of electromagnetic induction. In your description include how the data is collected and analysed.

[6]

(b) **Fig. 5.3** shows the poles of a powerful electromagnet producing a uniform field in the gap between them. The dimension of each pole is 0.10 m by 0.080 m. There is no field outside the gap. A circular coil of 80 turns is placed so that it encloses the total flux of the magnetic field.

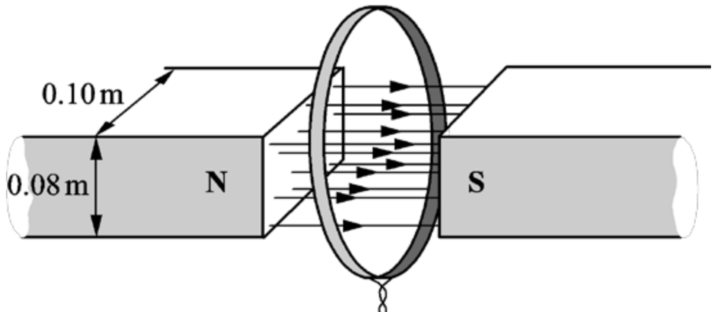

Fig. 5.3

i. The current in the electromagnet is reduced so that the field falls linearly from 0.20 T to zero in 5.0 s.

Calculate the initial flux in the gap and hence the e.m.f. generated in the coil during this time.

induced e.m.f. = ____________________ V **[2]**

ii. The coil is part of a circuit of total resistance *R* so that a current is generated in the circuit while the field is collapsing.

Draw on the coil in **Fig. 5.3** the direction of this induced current.

State how you applied the laws of electromagnetic induction to deduce the direction of this current.

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____ **[2]**

11(a) An astronomer uses a spectrometer and diffraction grating to view a hydrogen emission spectrum from a star. The light is incident normally on the grating.

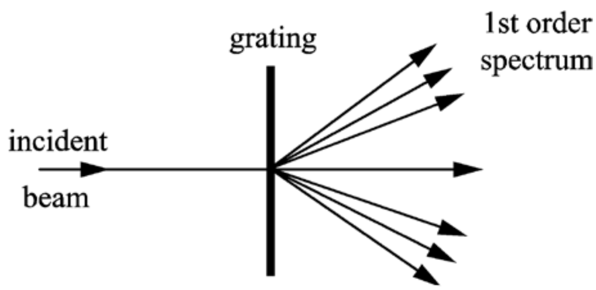


Fig. 6.1

First order diffraction maxima are observed at angles of 12.5°, 14.0° and 19.0° to the direction of the incident light as shown in **Fig. 6.1.**

Two of the wavelengths are 4.33×10^{-7} m and 4.84×10^{-7} m.

Calculate the wavelength of the third line.

wavelength = ____________________ m **[2]**

(b) In order to increase the accuracy of the values for wavelength, the student decides to look for higher order diffraction maxima.

i. State how this increases the accuracy.

____

____ **[1]**

ii. Calculate how many orders *n* can be observed for the shorter wavelength given in **(a)**.

n = ____________________ **[2]**

(c) These three emission lines all arise from transitions to the same final energy level. The part of the energy level diagram of hydrogen relevant to these transitions is shown in **Fig. 6.2**.

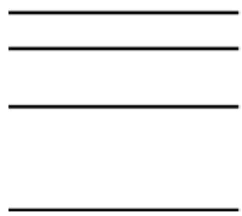


Fig. 6.2

i. Draw lines between the energy levels to indicate the transitions which cause the three emission lines and label them with their wavelengths.

[1]

ii. There are other possible transitions between the energy levels shown in **Fig. 6.2**. The least energetic of these produces photons of 4.8×10^{-20} J.

Calculate the wavelength of these photons.

State in which region of the electromagnetic spectrum this wavelength is found.

wavelength ____________________ m

region: ____

[3]

END OF QUESTION PAPER

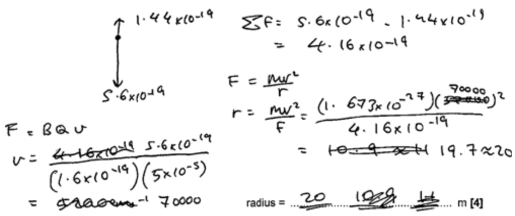
Question			Answer/Indicative content	Marks	Guidance
1	a	i	<u>Method 1</u> $d = 8.5 – 3.2$ (= 5.3(cm)) ($F = kd$ so) $F = 0.62 \times 5.30$ (= 3.3 (N)) $a = \frac{F}{m}$ so $a = \frac{3.3}{0.20}$ a = 17(m s^{-2}) <u>Method 2</u> ($F = kd$ so) $F = 0.62 \times 8.50$ (= 5.27(N)) $F_R = (0.62 \times 8.50) – (0.20 \times 9.81)$ (= 3.3 (N)) $a = \frac{F}{m}$ so $a = \frac{3.3}{0.20}$ <u>Method 3</u> ($F = kd$ so) $F = 0.62 \times 8.5$ (= 5.27(N)) $\left(a = \frac{F}{m} \text{ so}\right) a = \frac{5.27}{0.20}$ = 26 (m s$^{–2}$) $a_{initial}$ = 26.35 – 9.81 = 17 (m s$^{–2}$)	C1 C1 A1 (C1) (C1) (A1) (C1) (C1) (A1)	**Mark whichever method leads to the most marks** d = 5.3cm does not need to be calculated explicitly but seeing 5.3 implies first C1 mark F = 3.3N does not need to be calculated explicitly but seeing 3.3 implies both C1 marks **Allow** k = 0.61 (N cm^{-1}) leading to F = 3.2 (N)... ... and a = 16 (m s$^{–2}$) F = 5.27(N) does not need to be calculated explicitly but seeing 5.27 or 5.3 implies first C1 mark **Allow** k = 0.61 (N cm^{-1}) leading to F = 5.19 (N) F = 3.3(N) does not need to be calculated explicitly but seeing 3.3 implies both C1 marks **Allow** k = 0.61 (N cm^{-1}) leading to F = 3.2 (N)... ... and a = 16 (m s$^{–2}$) F = 5.27(N) does not need to be calculated explicitly but seeing 5.27 or 5.3 implies first C1 mark **Allow** k = 0.61 (N cm^{-1}) leading to F = 5.19 (N) Note: a= 26 (ms$^{–2}$) is an intermediate calculation for $a_{initial}$ in this method only and is **not** the A1 mark **Allow** k = 0.61 leading to F = 5.19 and a = 16 <u>Examiner's Comments</u> There are two measurements for extension given here: an extension of 3.2cm under a load of (0.2 × 9.81) N and an extension of 8.5cm under a force of F + (0.2 × 9.81) The easiest way to approach the question is to recognise that the extension due to F alone must be (8.5 – 3.2) = 5.3cm. $F = kx$ then becomes F = 0.62 × 5.3 (since k is given in N cm^{-1}) and so we can use $F = ma$ with m = 0.2kg to find the acceleration a.

Question			Answer/Indicative content	Marks	Guidance
					However, other valid methods were also given credit.
		ii	Use of $a = (–)\omega^2 x$ Use of $f = \dfrac{\omega}{2\pi}$ $f = 2.8$ (Hz) <u>Alternative method:</u> ($a = F / m = kx/ m$ and $a = (-)\ (–)\omega^2 x$ gives) $\omega^2 = \dfrac{k}{\mathrm{m}}$ $f = \dfrac{1}{2\pi}\sqrt{\dfrac{k}{m}}$ $= \dfrac{1}{2\pi}\sqrt{\dfrac{0.62 \times 100}{0.2}}$ $f = 2.8$ (Hz)	C1 C1 A1 (C1) (C1) (A1)	The two C1 marks are independent and XP in one does not imply XP in the other **Not** just formula alone Expect $a = 17$ but **allow** ECF of a from **(b)(i)** **Allow** any value for x **Not** just formula alone Use of $a = (–)(2\pi f)^2 x$ scores both C1 marks **Allow** $f = 2.9$ (Hz) **Allow** $T = 2\pi\sqrt{\dfrac{m}{k}}$ $f = \dfrac{1}{T}$ **Allow** $f = 2.9$ (Hz) <u>Examiner's Comments</u> The answer to this question is frequency = 2.8Hz, since ω depends only on k and m ($\omega^2 = k/m$). However, most candidates used the formula $a = (-)\omega^2 x$ together with appropriate values for a and x.

Question			Answer/Indicative content	Marks	Guidance
	b	i	(motion of magnet M causes) a change of flux (linkage) in coil Y (inducing an e.m.f.) there is an (induced) <u>current</u> in (or through) coil X alternating current / field / flux in coil X interacts with the field of magnet L (causing an alternating force)	B1 B1 B1	**Allow** field (lines of M) cuts (turns of) coil Y **Allow** the coil or Y or solenoid for coil Y **Allow** the coils or the wire(s) or X for coil X **Ignore** (induced) e.m.f. **Not** changing or varying or oscillating for alternating **Allow** current / field / flux in coil X interacts with field of magnet L **to** cause an alternating force **Allow** changing direction for alternating **Allow** combines for interacts **Allow** cuts across for interacts with <u>**Examiner's Comments**</u> Clarity in explanation was important here, as there are two magnets, M and L, plus two coils, X and Y. It is a change in flux linkage in coil *Y* which leads to an induced alternating current in coil *X*. This current creates an alternating magnetic field in coil *X* which interacts with the field of magnet *L* to create an alternating force on L. **Assessment for learning** Many explanations were too generalised: 'Faraday's Law states that there must be an induced emf which is proportional to the rate of change of flux linkage' or 'Fleming's left hand rule states there must be a force on the magnet' were often seen. Candidates should be encouraged to write in less general terms and to focus their answer on the specific question.

Question			Answer/Indicative content	Marks	Guidance
		ii	frequency of magnet L (always) equals (forcing/driving) frequency of vibration generator / magnet M resonance occurs at / close to 2.5 Hz amplitude is maximum at resonance	B1 B1 B1	**Allow** frequency of magnet increases with frequency of vibration generator May be seen from a **labelled** graph of amplitude against frequency **Allow** resonance occurs when forcing / driving frequency = natural frequency May be seen from a labelled graph of amplitude against frequency <u>Examiner's Comments</u> Many candidates did not realise that this was a question about resonance, presumably because of the unfamiliar context of the question. Common problems in 4(c)(ii) • not answering every part of the question: most candidates forgot to describe how the frequency varied as well as the amplitude • not realising that the vibration generator is driving the oscillation of L, and that this is a question about resonance • not labelling the scales on the graph of amplitude against frequency (or just using letters such as *A* and *f*) • failing to mark the resonance frequency as 2.5Hz (instead calling it f_o)
			Total	12	

Question			Answer/Indicative content	Marks	Guidance
2	a		F (= EQ) = 0.90 × 1.60 × 10^{-19} = 1.4(4) × 10^{-19} (N)	B1	Working and answer must both be shown Answer must be given to 2sf or more Unit need not be given but, if given, must be correct Examiner's Comments This was an easy introduction to the question, which used the definition of electric field strength; $E = F_E / Q$. Being a 'show that' question, candidates needed to show their working in full, including writing the value for the electronic charge (rather than simply 'e') and giving the answer to at least 2 s.f.
	b	i	($F = BQv$ but B and Q are constant, so) (the magnitude of) the velocity is different /changes	B1	**Allow** speed **Ignore** the direction is different Examiner's Comments The force on a charged particle moving at right angles to a magnetic field is given by the formula $F_{mag} = BQv$. Since B and Q are constants in this case, the reason for the different magnitude of F must be that the proton has a different velocity, v. **Common problems in 6(b)(i)** • using the formula $F = BIl\sin\theta$ and suggesting that the proton might be travelling at a different angle to the field, not realising that the proton is always travelling at right angles to the magnetic field in this question • suggesting that the proton may be in a weaker (or stronger) field at X than at Y, not realising that the magnetic field is uniform and so its field strength is constant throughout
		ii	$v = (\frac{F_{mag}}{BQ} =) \frac{5.6 \times 10^{-19}}{5.0 \times 10^{-5} \times 1.60 \times 10^{-19}}$ resultant force F_R= (5.6 – 1.4) × 10^{-19} $r = \left(\frac{mv^2}{F_R} =\right) \frac{1.673 \times 10^{-27} \times (7.0 \times 10^4)^2}{4.2 \times 10^{-19}}$ r = 20 (m) Alternative all-in-one method:	C1 C1 C1 A1 (C1) (C1) (C1) (A1)	v = 7.0 × 10^4 (m s^{-1}) implies first C1 **Allow** 10^{-19} for 1.4×10^{-19} (giving F_R= 4.6 × 10^{-19}) F_R = 4.2 × 10^{-19} implies second C1 Do not credit if used as F_{mag} in F_{mag} in $F_{mag} = BQv$ Third C1 is for correct substitution into formula **Allow** m_p = 1.67 × 10^{-27} kg given to 3 s.f. **Not** m_p = 1.661 × 10^{-27} kg or m_p = 1.675 × 10^{-27} kg **Allow ECF** for incorrect v

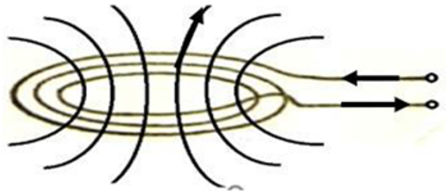
Question			Answer/Indicative content	Marks	Guidance
			$r = \frac{m{F_{mag}}^2}{F_R\, B^2 Q^2}$ resultant force $F_R = (5.6 - 1.4) \times 10^{-19}$ $r = \frac{1.673 \times 10^{-27} \times (5.6 \times 10^{-19})^2}{4.2 \times 10^{-19} \times (5.0 \times 10^{-5})^2 \times (1.60 \times 10^{-19})^2}$ r = 20 (m)		Use of $F_R = 5.6 \times 10^{-19}$ or $= 1.4 \times 10^{-19}$ is **XP** **Allow** r = 19 (m) F_R = **4.2 × 10**$^{-19}$ (4.16×10^{-19} to 3sf) implies second C1 An incorrect value of F_R is **XP** from this point Third C1 is for correct substitution into formula **Allow** r = 19 (m) <u>Examiner's Comments</u> This question could not be done in one step, by equating the magnetic force to the centripetal force. This is because, at X, the centripetal force is being provided by a combination of forces from both the electric and the magnetic field. The easiest approach is to find the velocity of the proton using $F_{mag} = BQv$ (the value for F_{mag} is given in the diagram as 5.6 × 10^{19} N). This velocity v can then be used in the formula $F = mv^2/r$ in order to calculate the radius, r. F here is the *resultant* force towards the centre of the circle, which is found from magnetic force downwards - electric force upwards (the electric force having been calculated in part (a)). Exemplar 3 is an example of a correct answer, clearly written to show each stage in the calculation: Exemplar 3 	

Question			Answer/Indicative content	Marks	Guidance
		iii	$\lvert\text{resultant force}\rvert = (\sqrt{3.9^2 + 1.4^2}) \times 10^{-19}$ $\lvert$resultant force$\rvert$ = 4.1 × 10^{-19} (N)	C1 A1	**Ignore** attempt to calculate weight of proton **Allow** $F_E = 10^{-19}$ **Allow** $\lvert F\rvert$ = 4.0 × 10^{-19} (N) using F_E = 1.0 × 10^{-19} **Allow** $\lvert F\rvert$ = 4.2 × 10^{-19} (N) using F_E = 1.44 × 10^{-19} <u>Examiner's Comments</u> There are two forces acting on the proton at Y: an electric force upwards (given in (a)) and a magnetic force to the left (shown on the diagram). These two forces act at right angles to each other, and so the magnitude of their resultant can be found using Pythagoras's Theorem. Credit was given for using a value for the electric force to 1, 2 or more significant figures.
		iv	<u>resultant / net</u> force is not perpendicular to velocity work is done on proton (therefore kinetic energy changes so speed is not constant)	B1 B1	**Allow** direction of motion / path but **not** speed for velocity **Allow** acceleration / <u>resultant</u> force is not (always) towards centre (of circle) **Allow** electric force is not perpendicular to velocity / is in the same direction as velocity **Ignore** references to centripetal **Ignore** references to centripetal <u>Examiner's Comments</u> At Y, the proton is moving downwards, with a resultant force being the combination of an electric force upwards and a magnetic force to the left (calculated in part 1). The resultant force cannot be at right angles to the velocity, so we cannot have circular motion. The component of the resultant force acting in the direction of the proton's motion will do work on the proton and change its speed. So, the proton cannot be travelling at a constant speed.
			Total	**10**	

Question			Answer/Indicative content	Marks	Guidance
3			***Level 3 (5–6 marks)** Detailed method **and** analysis which clearly distinguishes between gamma, beta-plus and beta-minus *There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.* **Level 2 (3–4 marks)** Some method **and** analysis which clearly distinguishes between any two of the sources *There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence.* **Level 1 (1–2 marks)** Limited method **or** limited analysis *There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.* **0 marks** *No response or no response worthy of credit.*	B1 × 6	Use level of response annotations in RM Assessor **Indicative scientific points may include:** **Method** • Measure background count and subtract from source count • Clamp source pointing away from you • Safety precautions (handle source with tongs, limit time etc.) • Record count over fixed time period • Investigate variation of count rate with range • Place aluminium sheets between source and radiation counter • Set up magnetic field at right angles to emission direction in order to investigate deflection of charged particles • Move radiation counter to find direction of deflection **Analysis** • Gamma has longest range in air, beta minus and beta plus have similar range (or β^+ has shortest range due to annihilation in air) • Gamma penetrates aluminium <u>which is (more than a few mm) thick</u> whereas beta does not • Gamma is undeflected by magnet (because neutral) • Beta radiation is deflected by magnet (because charged particles) • Beta plus and beta minus are deflected in opposite directions • because they have opposite charges / beta plus particle is a positron and beta minus particle is an electron • Use Fleming's left-hand rule to determine charge on beta particle through the direction of its deflection • With beta-plus, current is in same direction as motion of particle (opposite for beta-minus) <u>Examiner's Comments</u> When marking a LoR question, the examiner first decides the level of response (1, 2 or 3) by considering the quality of the physics. The examiner then

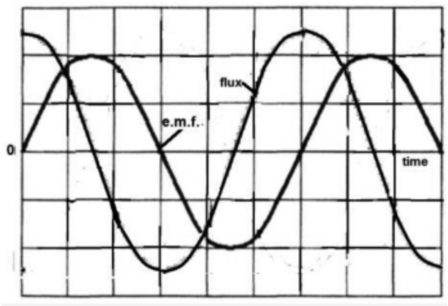
Question				Answer/Indicative content	Marks	Guidance
						decides whether to award the top of the level (for a well-structured and relevant response) or the bottom of the level (for a poorly structured and mostly irrelevant response). In this LoR question, responses in Level 1 often contained insufficient key physics. The experiment described was extremely safe and painstaking but, at the end of the day, the three sources were not clearly distinguishable. Low level responses frequently included instructions on how to test for an alpha source. The magnet was often used either to pick up the sources or to attract/repel charged particles. In a typical top level response, the thickness of aluminium used to block beta (but not gamma) was specified. A perpendicular magnetic field was used to separate the charged particles and the direction of their travel was predicted using an accurate diagram and Fleming's left hand rule. There were three possible approaches: 1. Investigate the range in air. Gamma has the largest range and beta-plus the smallest (due to annihilation). (The answer 'both beta-plus and beta-minus have a similar range in air' was also accepted). 2. Investigate the path through a perpendicular magnetic field. Gamma is undeflected; beta-plus and beta-minus would deflect in opposite directions. 3. Investigate the penetrative power. Since the materials and thicknesses available were not specified, it was easiest to keep increasing the thickness and density of the materials until all but one of the sources was blocked, thus identifying the gamma source. Many candidates were unable to distinguish clearly between the two beta sources because they did not understand how beta-plus and beta-minus particles would travel in a magnetic field. Many candidates erroneously thought that a magnet had a positive pole (which would attract beta-minus) and a negative pole (which would attract beta-plus).

Question			Answer/Indicative content	Marks	Guidance
					Misconceptions • A bar magnet does not have positive and negative poles – it has north and south poles • A magnet does not attract or repel charges – it exerts a force on charged particles moving at right angles to its magnetic field
			Total	6	

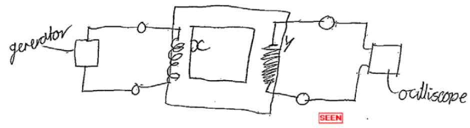
Question			Answer/Indicative content	Marks	Guidance
4		i		B1 B1	One correct line (or dot and cross) drawn Line must go through centre of coil **Allow** an incomplete line or a complete circle round the coil **Ignore** direction of arrow More than one line drawn All lines drawn must go through centre of coil and follow correct shape and <u>direction</u> of field **Ignore** spacing of lines **Ignore** any lines to the right of the coil
		ii	(the magnetic) flux (of the coil) links the <u>base</u> / <u>saucepan</u> (the size/direction of) the flux linkage (constantly) <u>changes/alternates</u> (causing an alternating induced e.m.f.) (induced) <u>current</u> is large because metal/base/ saucepan has low resistance	B1 x 2	2 out of 3 possible marking points **Allow** (the magnetic) field lines cut the (base of the) <u>saucepan</u> **Allow** the (magnetic) field constantly changes/alternates **Allow** a bald statement of Faraday's Law
		iii	The resistance of glass-ceramic/the (cook"s) hand is (very) large So (induced) <u>current</u> (or heating effect of <u>current</u>) is zero/negligible	M1 A1	**Allow** glass-ceramic/hand is an insulator/not a (good) conductor **Do not allow** the induced <u>e.m.f</u>. is (very) small
			Total	**6**	

Question			Answer/Indicative content	Marks	Guidance
5			◦ (Induced) e.m.f. is caused by a change in (magnetic) flux (linkage) / (Induced) e.m.f. is proportional (or equal to) the rate of change of (magnetic) flux (linkage) • The peaks are inverse / e.m.f. changes from positive to negative because: the rate of change of magnetic flux linking the coil changes sign **or** the flux (linkage) increases and then decreases **or** description in terms of Lenz's law as seen by coil to conserve energy • The e.m.f.becomes zero because: the (rate of) change of magnetic flux is zero when the magnet is in the middle of the coil • The second peak has a larger negative amplitude because: the **rate** of change of flux linkage is greater (when the magnet leaves the coil compared to when it enters) • The pulses have different widths because: the second Δt is shorter (since magnet accelerates) **or** areas under curves must be the same (because total change of flux linkage is the same on entering and leaving coil) / area under curve = $V\Delta t$ = $N\Delta\phi$ (so bigger V leads to smaller Δt)	B1 x 3	**Maximum 3** marks from 4 marking points. **Not** voltage or p.d. or current for e.m.f. **Accept** 'cutting of field lines by coil' for 'change in flux' Answers to any of the last three points must link clearly to the correct graph characteristic **Allow** the North (or South) pole first approaches then recedes **Ignore** magnet approaches then recedes / field increases then decreases **Not** torch is inverted **Allow** no field lines are being cut **Allow** the magnet is accelerating / is travelling faster when it exits the coil **Examiner's Comments** Candidates need to remember to look at the command word in the question. Here it was 'explain'; not 'describe'. The key features to be explained were: • why is an e.m.f generated? • why does the e.m.f change sign? • why does the e.m.f fall to zero halfway through the fall? • why is the maximum negative e.m.f greater than the maximum positive e.m.f / why is the width of the second peak smaller than that of the first peak? The strongest responses were those where candidates stated at the outset what gave rise to the e.m.f. Some candidates clearly recognised the need to state Faraday's law, but simply quoted the

Question			Answer/Indicative content	Marks	Guidance
					formula without defining any terms and so could not receive credit. Weaker responses were characterised by describing the shape of the graph in terms of the position of the magnet - often incorrectly – rather than in terms of flux linkage. A common misconception was stating that the negative peak was caused by the magnet returning after an inversion, with the zero e.m.f. just after 0.1s being caused by the magnet being temporarily stationary. However, the question clearly states that ‘Fig.3.3 shows … the e.m.f. generated … as the magnet falls the distance h’. Exemplar 3 below demonstrates clearly this common misconception. **Exemplar 3** The first part of the curve the Emf is increasesing as the magnet gets closer to the coil. The peak is when the magnet is in the middle of the coil as the magnet moves away so does the curve decrease. As the magnet is now moving in the opposite direction the emf is negative. It takes less time on the way back as the electric field repels it. [3]
			Total	3	

Question			Answer/Indicative content	Marks	Guidance
6	a	i	the flux in the coil changes/ increases/ decreases/ varies (caused by the spinning/rotating magnet) causing a sinusoidal/alternating e.m.f./AW	B1 B1	**or** e.m.f. is proportional to /equals rate of change of flux linkage/linking the coil **or qualification, e.g.** magnet vertical gives minimum flux through core or maximum rate of change of flux or vice versa with magnet horizontal **or** maximum flux is when emf is zero **or** minimum flux is when emf is maximum **or** vice versa
		ii		B1	**allow** ± cos wave of correct period, constant amplitude at least one cycle **N.B.** quality: curve must look like a reasonable sine wave as one is present on the page to copy Examiner's Comments In part (i) many of the candidates described the phase shift that they drew in the sketch graph of part (ii) by stating either the magnitude or the rate of change of the flux linkage when the induced e.m.f. was zero or a maximum. The majority quoted Faraday's law either in words or as a mathematical equation. Some candidates introduced current and Lenz's law not appreciating that an oscilloscope is effectively a voltmeter. Few described the whole picture of a steadily rotating magnetic field sweeping through a coil creating a changing flux linkage.
		iii	$\phi = BA = V/2\pi fN = 1.2/(2 \times \pi \times 24 \times 150)$ $\phi = 5.3 \times 10^{-5}$ Wb / T m^2	 B1 B1	 **allow no other** unit combinations; NOT T m^{-2}

Question			Answer/Indicative content	Marks	Guidance
	b		**Level 3 (5–6 marks)** Clear description, some measurements and full analysis *There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.* **Level 2 (3–4 marks)** Some description, some measurements and some analysis. *There is a line of reasoning presented with some structure. The information presented is in the most-part relevant and supported by some evidence.* **Level 1 (1–2 marks)** Limited description and/or limited measurements and/or limited analysis *There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.* **0 marks** No response or no response worthy of credit.	B1 × 6	**Indicative scientific points may include:** **Description** **a.** Signal generator/a.c. supply connected to coil X **b.** Coil Y connected to voltmeter / oscilloscope (can be ondiagram) **c.** Use oscilloscope to determine period / frequency or readoff signal generator **d.** Adjust signal generator / use of rheostat to keep currentconstant in coil X **Measurements** **1.** Vary *f* and measure *V* **2.** Keep <u>current</u> in coil X <u>constant</u> **3.** Detail on how to measure e.m.f. e.g. 'height x *y*-gain' **4.** Detail on how to measure period on oscilloscopescreen using time base and hence *f* **Analysis** **1.** Determine f from period measurement, f = 1/T **2.** Plot a graph of *V* against *f* **3.** Relationship valid if straight line through the origin <u>Examiner's Comments</u> From the proposed arrangements for the investigation, it was apparent that most of the candidates were unfamiliar with the most suitable equipment for this experiment, namely a signal generator. Many improvised by using an ac supply with a variable frequency. A minority of these believed that by increasing the voltage of their power supply it would alter the frequency. Most drew a cell or battery symbol for the ac supply. Others improvised by using the rotating magnet from part (a) but had not realised the significance of the calculation in part (a)(iii) which indicated that at 24 revolutions per second the output voltage was 1.2 V. This made the suggested method of using a stop watch to find the period of rotation impracticable. Few realised that the oscilloscope as a voltmeter could measure both the output voltage and the period of the ac. The instrument was often

Question			Answer/Indicative content	Marks	Guidance
					connected in series in the primary circuit. No one realised that the input current has to be constant to provide a constant flux. Despite all of these difficulties most candidates managed to write sensible statements worthy of credit but rarely full marks. The author of the example shown (exemplar 9) has used the rotating magnet as the ac source and continued with the clues from part (a) to produce an L3 quality answer. **Exemplar 9**  Using some form of ~~signal generato~~ alternating current generator with variable frequency ~~set~~ connected to X and an oscilloscope connected to Y Vary frequency f from the generator either by increasing ~~rmp~~ rotation of ~~motor~~ magnet or using a built in frequency adjustment. from the oscilloscope measure V by the ~~dis~~ height of the peaks X voltage base setting and frequency by distance by measuring time period as distance between peaks ~~ano~~ X time base setting and $f = \frac{1}{T}$ as $\frac{V_x}{V_y} = \frac{N_x}{N_y}$ and N_x and N_y (number of turns in X and Y) are constant $V_x \propto V_y$ so as V_x should be $\propto$ frequency ~~$V_y \propto f$~~ so $V_y \propto$ frequency so plotting a graph of V_y ~~and f~~ against f for varying frequencies should be a straight line through the origin if the relationship is correct. L3
			Total	11	

Question			Answer/Indicative content	Marks	Guidance
7	a		F = Bev **and** F = eE E = V / a **or** F = (eE) = eV / a Bev = eV / a giving V = Bva	B1 B1 B1	**allow** Q or q for e **Examiner's Comments** This was an exercise in writing basic definitions in algebraic form and then using them to derive a given equation. More than half of the candidates managed to gain full marks with less than one third scoring zero. The presentation was sometimes difficult to follow with the inclusion of unnecessary equations and deletions and the substitution of *d* for *a* in the last line.
	b	i	I = nAev; v = 60 × 10^{-3}/1.2 × 10^{23} × 1.6 × 10^{-19} × 5.0 × 0.2 x10^{-6} v = 3.1 (m s^{-1})	C1 C1 A1	**allow** any subject
		ii	V = 80 × 10^{-3} × 3.1 × 5.0 × 10^{-3} = 1.2 × 10^{-3} (V)	A1	**ecf (b)(i); allow** 1.2 mV; 1.3 × 10^{-3} (V) **Examiner's Comments** This exercise of choosing a formula, substituting values in correct units and evaluating was done well with about three quarters of the candidates gaining full marks.
	c	i	Hall probe only compares B-fields / AW **or** V will be too small / less than 1 mV so not easy to measure	B1	allow any sensible comment, e.g. how do you convert the measured V into a B value

Question			Answer/Indicative content	Marks	Guidance
		ii	find B using F = B*Il*; F is measured by weighing magnets (e.g. placed on top pan balance assuming wire is fixed); graph of F against *I* to find B*(l)* from gradient / AW; greatest uncertainty: measurement of *l* in B-field sensible reason / justification for choosing *l* **or** small masses	B1 B1 B1 B1 B1	**max** 4 of the 5 marking points **alt** measure F by adding small masses to wire to return it to zero current position **or** use readings of F at several *I* to find average F/*I*, etc. **or** measurement of small masses in alt. method, etc quantitative suggestion about % error i.e. *l* small (1 mm in 60) leading to large % uncertainty **or** difficulty in determining edge / end of B-field **Examiner's Comments** Most candidates did not refer back to (b)(ii), noting that the potential difference across the Hall probe would be very small making the probe an unsuitable instrument for measuring the magnetic flux density, *B.* However almost all were familiar with the experiment where the magnets are mounted on a top pan balance with a fixed wire carrying the current. Only a small number varied the current and plotted a graph to obtain a more accurate value of *B*. Also few appreciated that the edges of the field spread out making the length of wire in the field the least reliable measurement.
			Total	12	

Question			Answer/Indicative content	Marks	Guidance
8	a		**Level 3 (5–6 marks)** A good plan with discussion of sensitivity and measurements that need taking. Detailed description of analysis needed linked to robust conclusions and consideration of a fair test. extra points from sections may balance omissions from others *The ideas are well structured providing significant clarity in the communication of the science.* **Level 2 (3–4 marks)** A good plan possibly with mention of sensitivity. Measurements that need taking should be described. Analysis linked to conclusions and possibly consideration of a fair test. extra points from sections may balance omissions from others *There is partial structuring of the ideas with communication of the science generally clear.* **Level 1 (1–2 marks)** A plan with discussion of measurements that need taking. Description of analysis needed linked to a conclusion. *extra points from sections may balance omissions from others The ideas are poorly structured and impede the communication of the science.* **Level 0 (0 marks)** Insufficient relevant science.	B1	**plan P** • investigate one variable with the other fixed • oscilloscope time base can be off • do rough preliminary test over range of variable to check that there is a suitable variation in oscilloscope V • choose and fix f of *I* and value of other variable (M3); • measure e.m.f. V for 5 or 6 settings of variable from oscilloscope screen **sensitivity S** • magnitude of detected signal depends on rate of change of flux linkage / Faraday's law through search coil • so increases with f and B (N and A of search coil are fixed) • for large B use small *L* f changing N **or** large N if changing *L* **measurements M** • measure (maximum) e.m.f. V (using V/cm scale setting) on oscilloscope • measure peak to peak distance on graticule if time base not switched off • keep *L* fixed and adjust croc. clips to change N **or** keep N fixed and alter *L* (use ruler) **analysis A** • record table of V against *N* or *L* • plot graph of V against *N* or 1/ *L* **conclusions C** • straight line graph • through origin is expected • to validate given relationship **fair test F** • ensure that Slinky coils are uniformly spaced and not touching together anywhere • croc. clips make good contact at only one point on coil • plane of coil must be vertical **and** coaxial with Slinky
	b	i	Two closed loops linking primary coil	B1	lines not touching / crossing, both passing only through iron core

Question			Answer/Indicative content	Marks	Guidance
		ii	magnetic flux ϕ: because the loops of magnetic field (are continuous and) all pass (through the iron core) through each coil	B1	**allow** magnetic flux is the number of lines of the magnetic field if **(i)** is correct
		iii iii	for magnetic flux density: 3 turn coil as A is smallest OR for magnetic flux linkage: 5 turn coil as largest number of turns	B1 B1	Note: (iii) and (iv) can be answered in either order ϕ is same in each coil, $B = \phi/A$ OR ϕ is same in each coil, m.f.l. = ϕN
			Total	10	

Question			Answer/Indicative content	Marks	Guidance
9	a	i	F upwards between poles	B1	
		ii	$F = BIl = 0.032 \times 2.5 \times 0.06 = 4.8 \times 10^{-3}$ (N)	B1	
	b	i	$a = (–)\ 4\pi^2 f^2 x = 4 \times 9.87 \times 4900 \times 0.004$	C1	**allow** 774 (m s^{-2})
		i	a = 770 (m s^{-2})	A1	
		ii	**1** sketch showing one wavelength and 140 (Hz)	B1	both sketch and value required for 1 mark
		ii	**2** driving force is around nodal point / AW;	B1	**max** 3 of the 4 marking points
		ii	points either side of nodal point try to move in opposite directions when force in one direction / AW;	B1	
		ii	move magnet to antinodal point; ¼ of distance between clamps	B1	**not** increase current
	c	i	$f\ \alpha\ \sqrt{T}$ so $f = 70/\sqrt{2}$ = 49 or 50 Hz	B1	
		ii	**1** μ increases / goes up by 0.4%	B1	**allow** +0.4% NOT 0.4%
		ii	**2** 0.2%,	B1	**or** half of answer to **(ii)1**
		ii	f is lower because μ is bigger and μ is on the bottom of the formula	B1	**or** greater inertia present with same restoring force / other physical argument
			Total	**12**	

Question			Answer/Indicative content	Marks	Guidance
10	a		*** Level 3 (5–6 marks)** At least P1 and P2 M1, M2, M4 and M5 At least A2 and A3 At least C1 and C2 *There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.* **Level 2 (3–4 marks)** At least P1 M1, M4 and M2 or M5 At least A3 At least C1 *There is a line of reasoning presented with some structure. The information presented is in the most-part relevant and supported by some evidence.* **Level 1 (1–2 marks)** At least P1 At least M1 and M4 At least A3 At least C1 *The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear.* **0 marks** No response or no response worthy of credit.	B1	**plan P** 1. vary speed of rotation of magnet using motor control 2. expect to see amplitude of signal increase and period of waveform decrease 3. measure (maximum) e.m.f. V and period T for each setting from oscilloscope screen. **measurements M** 1. maximum e.m.f. 2. measured from peak to peak distance on graticule 3. and using V/cm scale setting 4. period of rotation 5. measured along t-axis of graticule 6. and using s/cm time base setting. **analysis A** 1. record table of V, T 2. and (calculate and record) f = 1/T 3. plot graph of V against f **conclusions C** 1. astraight line graph 2. through origin 3. is required to validate Faraday's law.
	b	i	flux = BA = 0.20 × 0.10 × 0.080 = 0.0016 (Wb)	B1	
		i	induced emf = NBA/t = 80 × 0.0016/5 = 0.026 (V)	B1	
		ii	Lenz's law indicates that current must try to maintain the field as it collapses **or** current must produce same field as magnet to try to maintain the field.	M1	
		ii	current is anticlockwise in coil as viewed from S pole.	A1	

Question			Answer/Indicative content	Marks	Guidance
			Total	10	

Question			Answer/Indicative content	Marks	Guidance
11	a		$\lambda_1 = d \sin 12.5 = 4.33 \times 10^{-7}$ m giving $1/d = 5 \times 10^5$ or $d = 2 \times 10^{-6}$	C1	or $\lambda_2 = d \sin 14.0 = 4.84 \times 10^{-7}$ (m)
			$\lambda_3 = \sin 19.0/5 \times 10^5 = 6.51 \times 10^{-7}$ (m) **or** $\lambda_1 = d \sin 12.5 = 4.33 \times 10^{-7}$ and $\lambda_3 = d \sin 19.0$ so $\lambda_3 = 4.33 \times 10^{-7} \sin 19.0/\sin 12.5 = 6.51 \times 10^{-7}$ (m)	A1	or use $\lambda_2 = d \sin 14.0 = 4.84 \times 10^{-7}$ m sin 19.0/sin 12.5 = 0.326/0.216 = 1.50
	b	i	the uncertainty in the measurement of angle is the same for all angles and the bigger the angle measured the smaller the % error	B1	
		ii	$n_{max} = d \sin 90$	C1	
		ii	$= 1/(5 \times 10^5 \times 4.33 \times 10^{-7}) = 4.6$ but n is an integer so n = 4	A1	
	c	i	3 downward arrows correctly labelled.	B1	longest being 4.33×10^{-7} (m)
		ii	$\Delta E = hc/\lambda$	C1	
		ii	$\lambda = 6.63 \times 10^{-34} \times 3 \times 10^8/ 4.8 \times 10^{-20} = 4.1(4) \times 10^{-6}$ (m)	A1	
		ii	region: infra red	B1	**allow** ecf if wavelength calculation incorrect.
			Total	**9**	